

3. Row 2 Column 1 = 0

∴ The condition that for each column having a leading entry, all positions below the leading entry position must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

∴ The condition that for each column having a leading entry, all positions above the leading entry position must contain 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 1 & -2 & -1 \end{bmatrix}$ is in rref.

By defn, the pivots are: ~~Row~~ Row 1, Column 1 and Row 2 Column 2

There are no pivots in Column 3. ∴ By defn, x_3 is a free variable.

The corresponding equations are: 1. $x_1 + 3x_3 = 4$ 2. $x_2 - 2x_3 = -1$

$$x_1 = 4 - 3x_3 \text{ and } x_2 = 2x_3 - 1$$

In parametric vector form, the solution set is: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$

$$+ \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$$

The intersection of the two planes is the line through p in the direction of v .

2. Write the general solution of $10x_1 - 3x_2 - 2x_3 = 7$ in parametric vector form, and relate the solution set to the one found in Example 2.

By the defn. of $Ax=b$, we

Solution: $x_1 \begin{bmatrix} 10 \end{bmatrix} + x_2 \begin{bmatrix} -3 \end{bmatrix} + x_3 \begin{bmatrix} -2 \end{bmatrix} = \begin{bmatrix} 7 \end{bmatrix}$

get $\begin{bmatrix} 10 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \end{bmatrix}$

By Theorem 3, the matrix equation $Ax=b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} 10 & -3 & -2 & 7 \end{bmatrix} \xrightarrow[\times 1/10]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & -0.3 & -0.2 & 0.7 \end{bmatrix}$$

From Example 2, it's easy to see that the augmented matrix is in rref form.